Dudhichua Opencast Mine

🏭 Introduction – Dudhichua Opencast Mine

The **Dudhichua Opencast Mine** is a large and highly mechanized coal mine operated by Northern Coalfields Limited, a subsidiary of Coal India Limited. It is one of the key opencast projects in the Singrauli coalfield and plays a major role in supplying coal to thermal power plants.

Mining operations began in the early 1980s, and the mine has since developed into a high-capacity project with an annual production of around 20–25 million tonnes. It is known as a **"basket linkage mine"**, meaning it supplies coal to multiple power plants, including those operated by NTPC Limited. The mine uses advanced technologies such as draglines, shovels, and dumper systems, making it an important case study for large-scale surface mining in India.

🌍 Geographical Overview – Dudhichua Opencast Mine

The Dudhichua mine is located in the **Singrauli coalfield**, situated in the Sonbhadra district of Uttar Pradesh, near the border of Madhya Pradesh. This region is one of India's most important coal-bearing areas.

📍 Location & Setting

- Lies in the **Moher sub-basin** of the Singrauli coalfield
- Approximate coordinates: **24.16°N latitude and 82.67°E longitude**
- Well connected by rail and road networks for coal transportation

🌄 Physical & Geological Features

- Part of the **Vindhyan plateau region**, characterized by undulating terrain
- Contains thick, gently dipping coal seams suitable for opencast mining
- Coal type: **sub-bituminous**, mainly used for thermal power generation

🌊 Nearby Natural Features

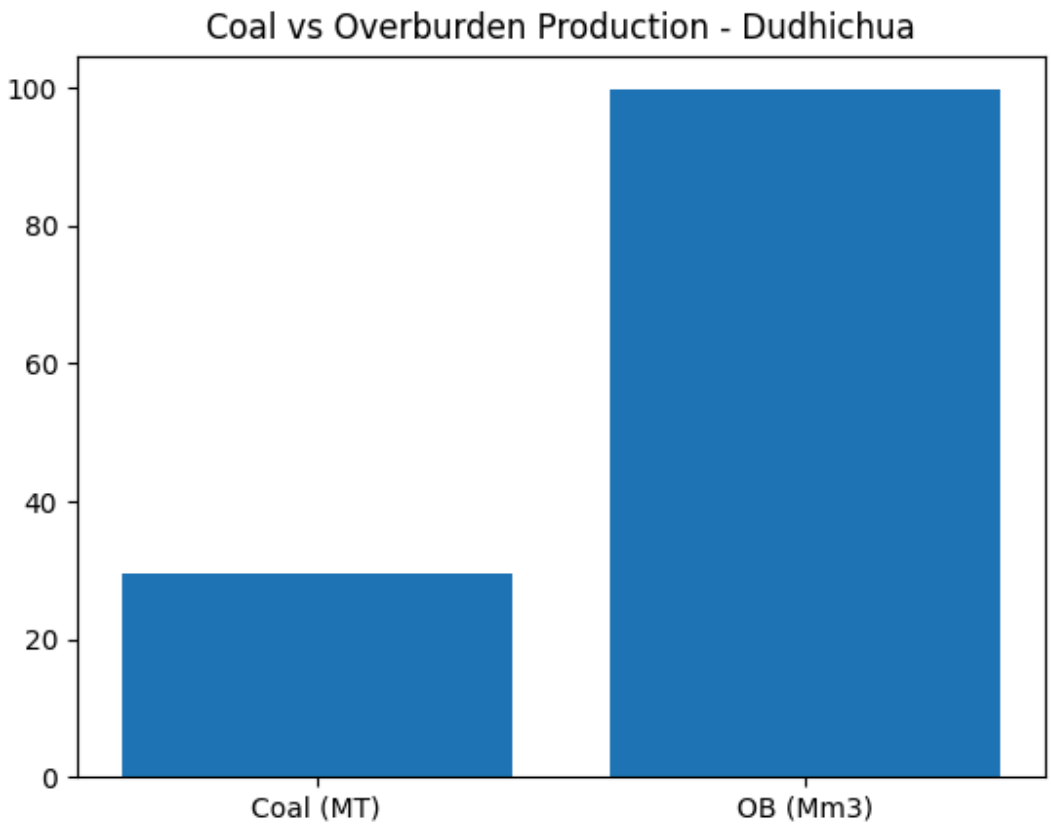
- Located close to the Rihand River and the Govind Ballabh Pant Sagar reservoir
- Availability of water supports mining and power plant operations

🌦️ Climate

- Tropical climate

- Very hot summers (up to ~48°C)
- Moderate to heavy monsoon rainfall
- Mild winters

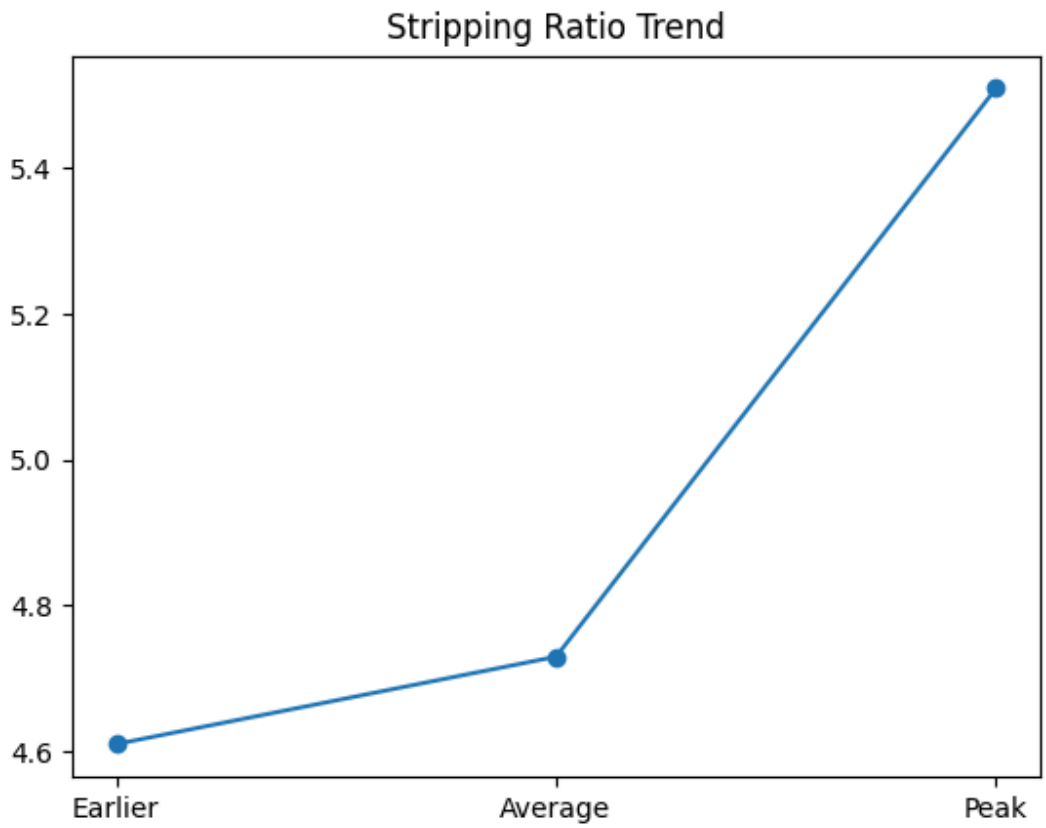
📊 1. Coal vs Overburden Production



📌 Description & Significance:

This chart compares coal output with overburden removal.
It shows that a large volume of waste must be removed before coal extraction.
Significance: Helps in calculating stripping ratio and determining economic viability of the mine.

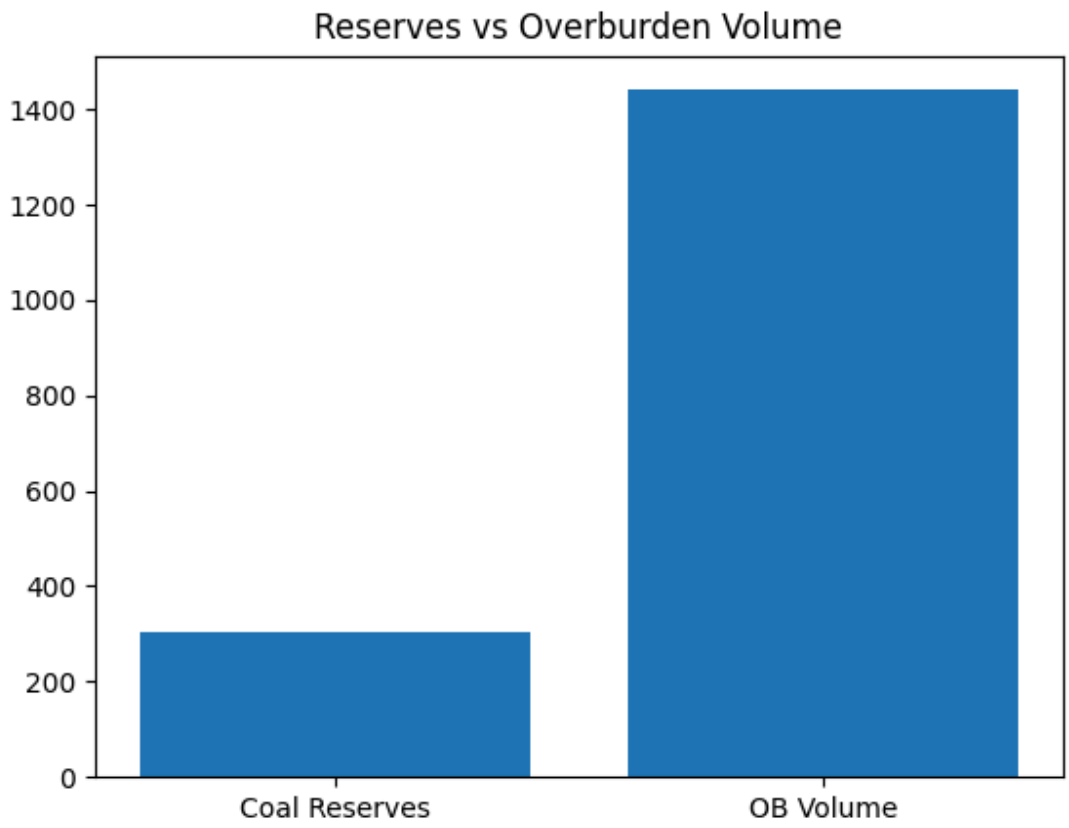
📊 2. Stripping Ratio Trend



📌 Description & Significance:

Shows increase in stripping ratio with depth.
Higher SR means more cost and more waste handling.
Significance: Critical for mine planning, cost estimation, and deciding mine life.

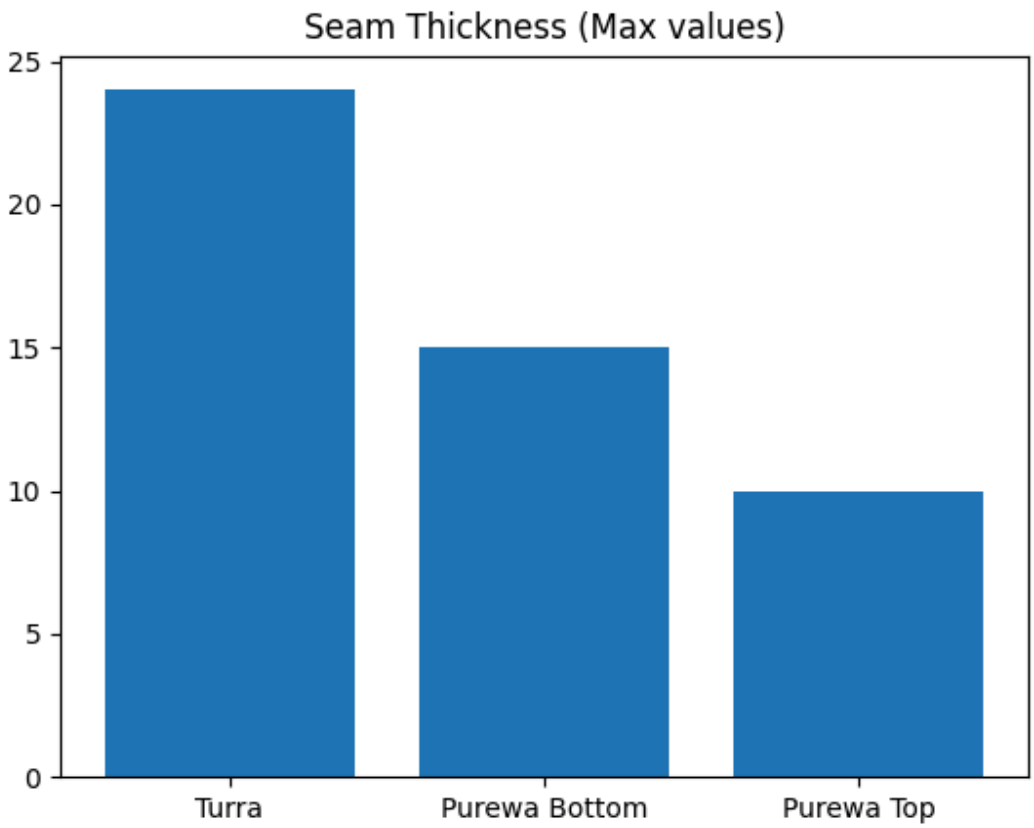
📊 3. Mineable Reserves vs OB Volume



📌 Description & Significance:

Shows proportion of useful mineral vs waste.
OB volume is much higher than coal reserves.
Significance: Important for dump design, land requirement, and equipment planning.

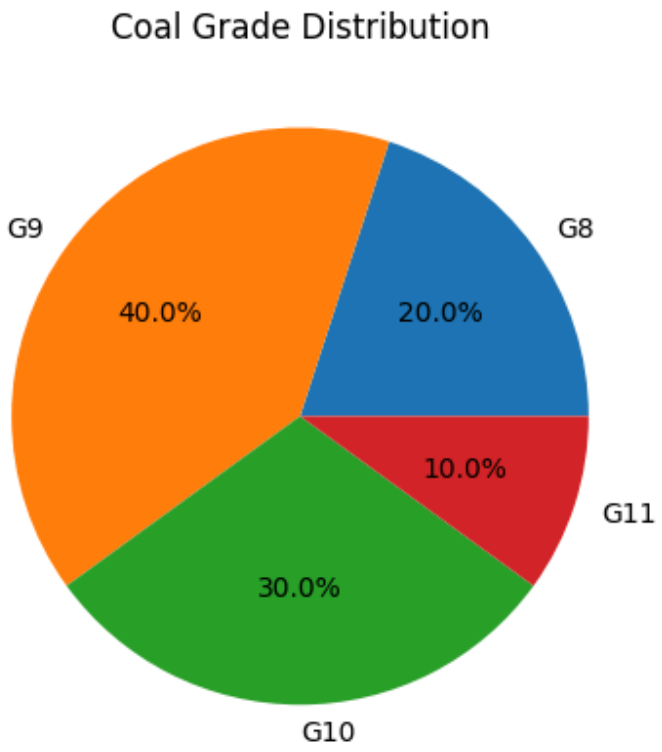
📊 4. Seam Thickness Distribution



📌 Description & Significance:

Shows variation in coal seam thickness.
Thicker seams are easier and more economical to mine.
Significance: Helps in selecting mining method and equipment size.

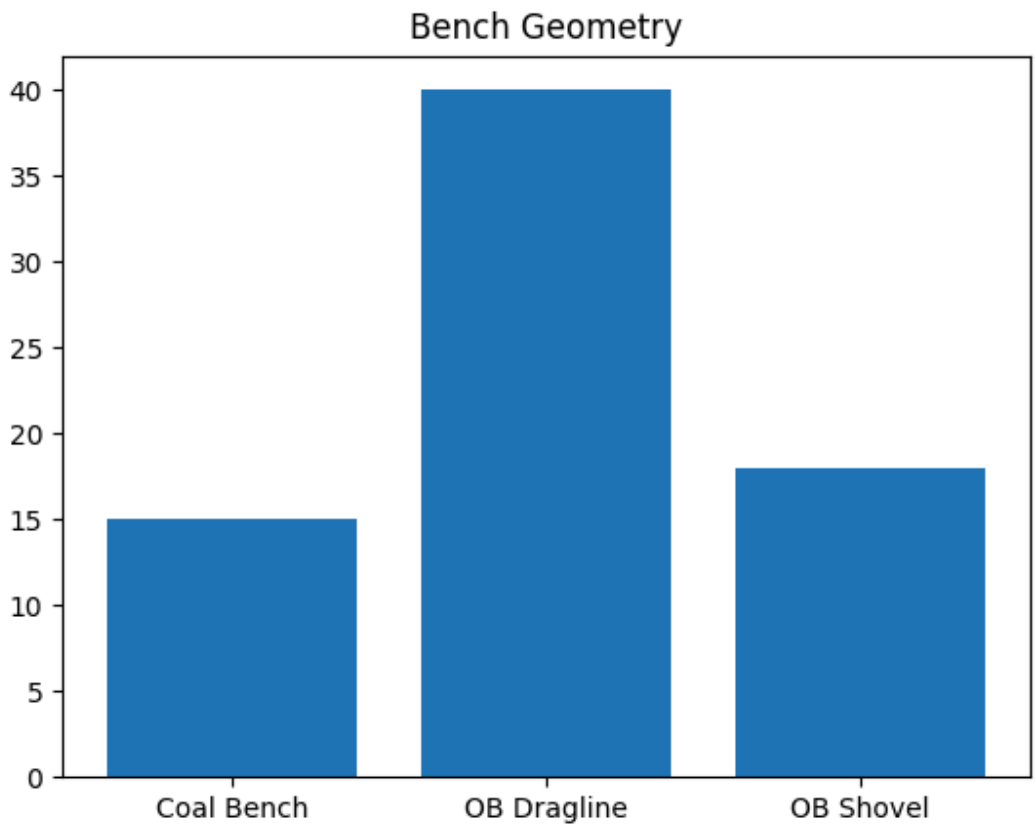
📊 5. Coal Grade Distribution



📌 Description & Significance:

Represents quality of coal based on calorific value.
Moderate grade coal is suitable for thermal power plants.
Significance: Affects pricing, blending, and plant efficiency.

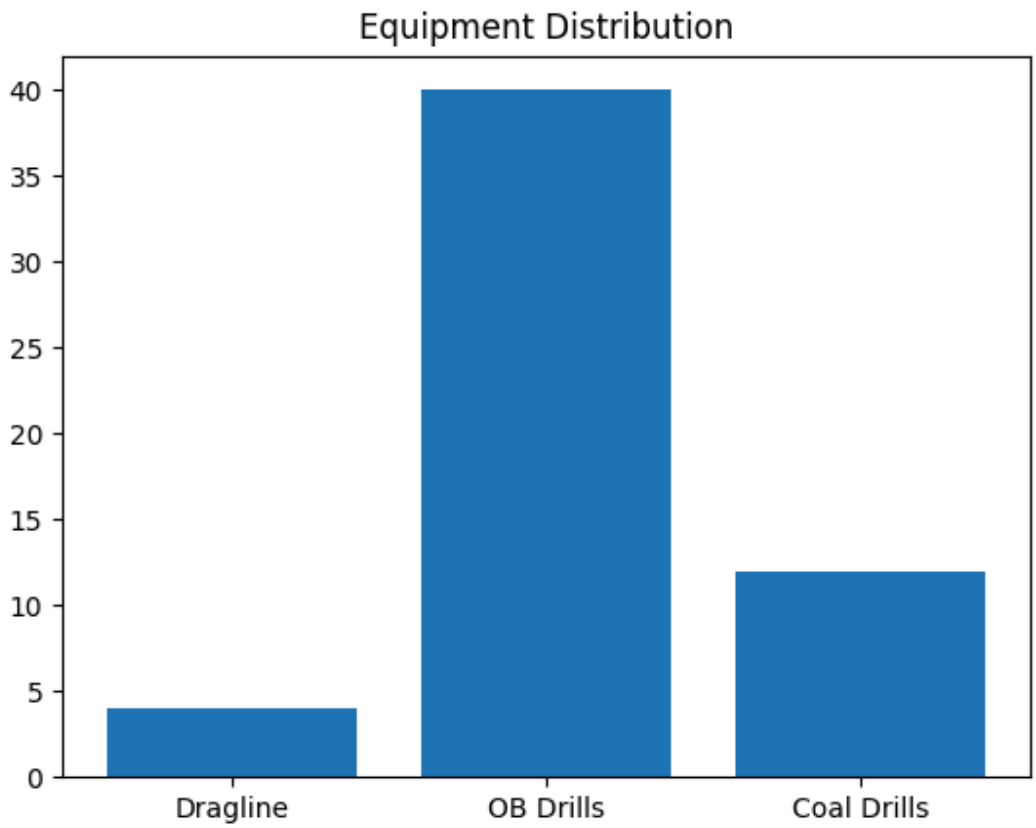
📊 6. Bench Geometry (Mining Design)



📌 Description & Significance:

Shows design of benches in the mine.
Different heights are used for different equipment.
Significance: Important for slope stability and safe mining operations.

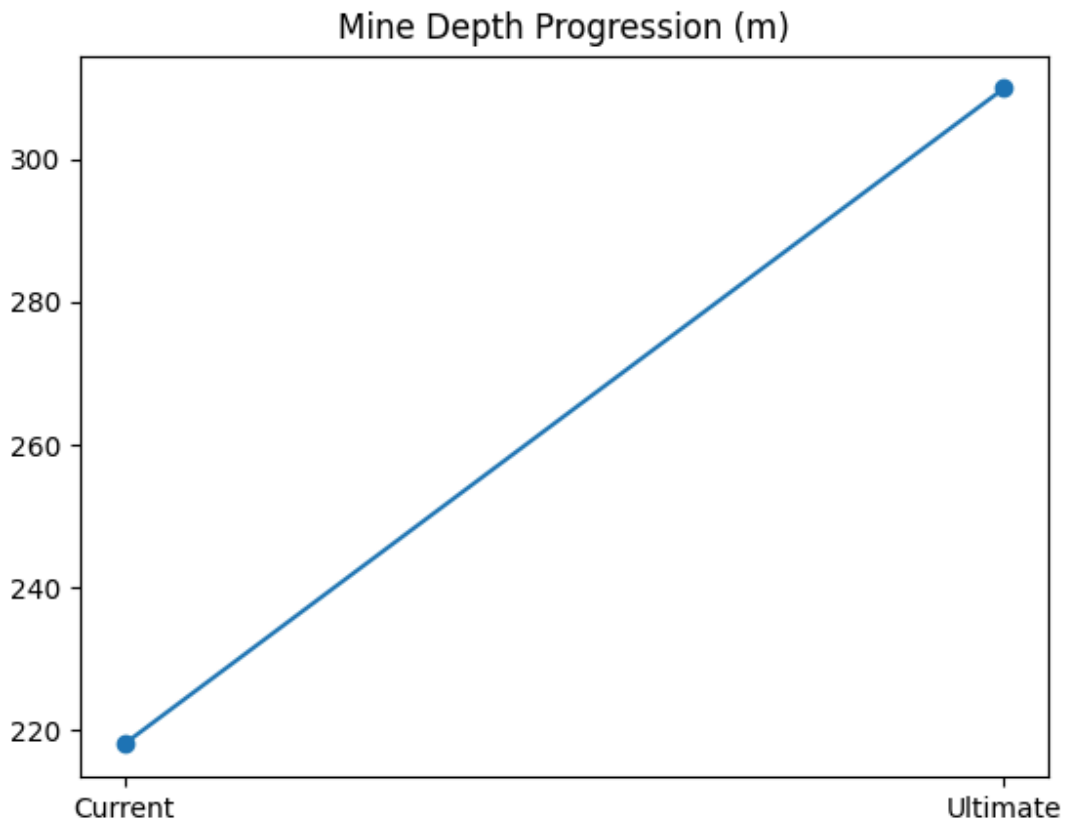
📊 7. Equipment Distribution



📌 Description & Significance:

Shows deployment of drilling equipment across mine sections.
More drills are required in OB due to higher volume.
Significance: Helps in equipment planning and productivity optimization.

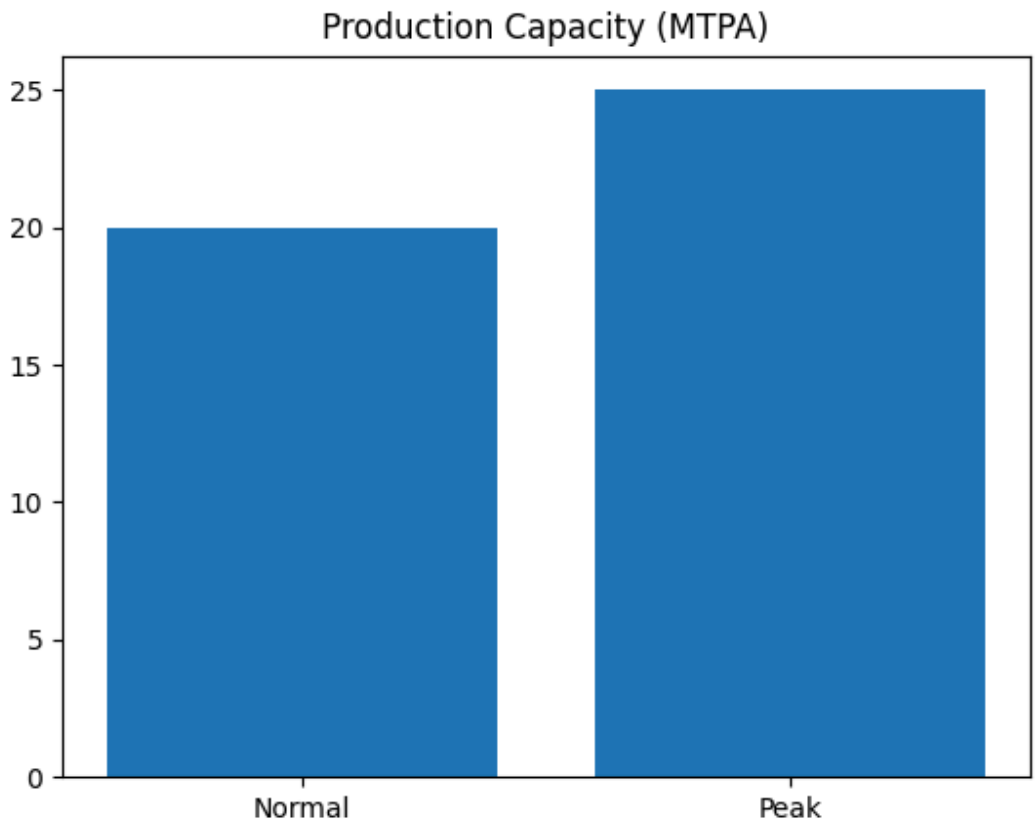
📊 8. Depth vs Mining Difficulty



📌 Description & Significance:

Shows how mining becomes more complex with depth.
Deeper mines require more OB removal and slope control.
Significance: Important for long-term mine planning and safety.

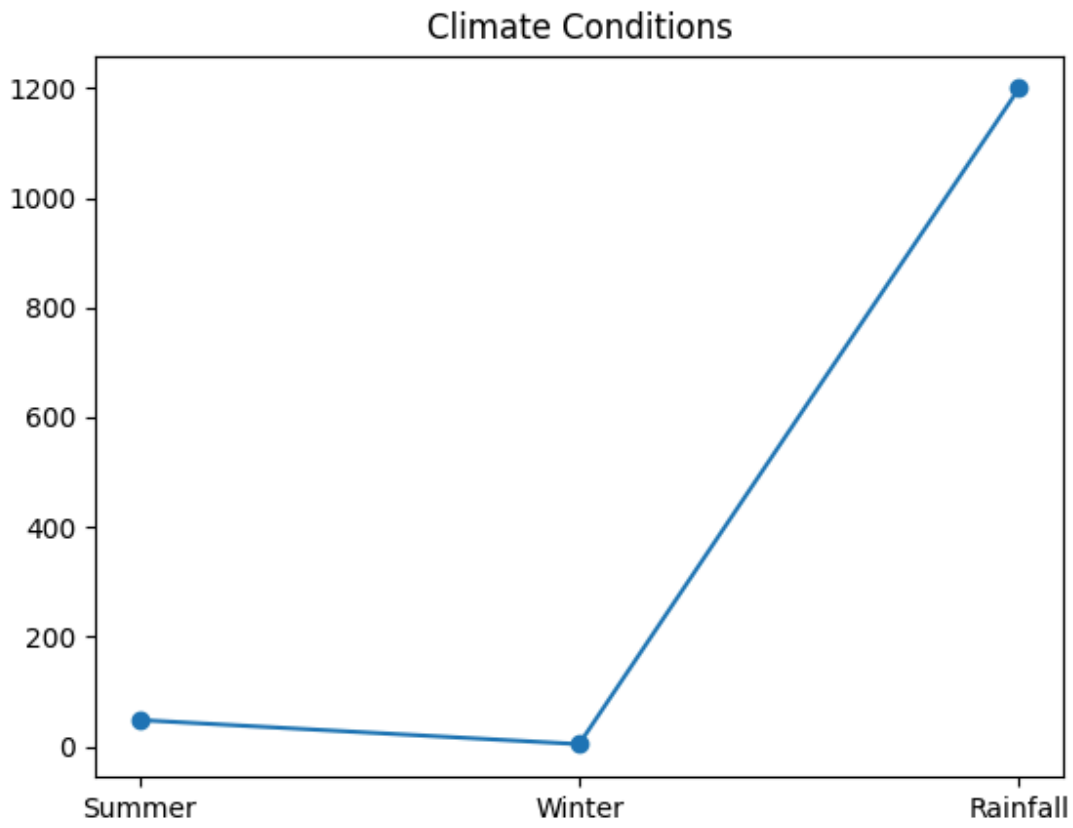
📊 9. Production Capacity Plan



📌 Description & Significance:

Shows designed vs peak production capacity.
Indicates expansion potential of the mine.
Significance: Helps in scheduling, investment planning, and equipment sizing.

📊 10. Temperature & Climate Impact



📌 Description & Significance:

Shows climatic conditions affecting mining operations.
Extreme heat and monsoon impact productivity.
Significance: Important for mine scheduling, safety, and drainage design.